

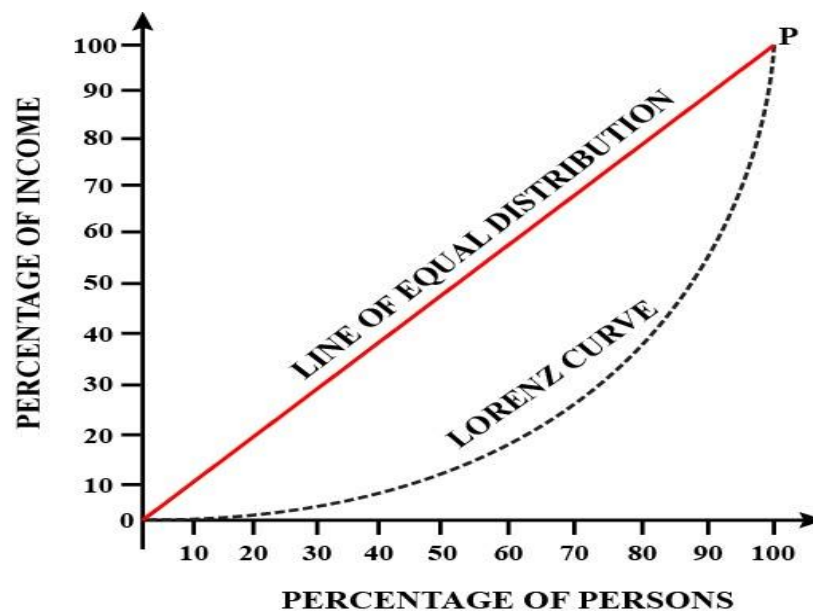
Practical Problem_3: Lorenz Curve_1

Draw a Lorenz curve from the data given below:

Income	Number of Persons
0	0
100	80
200	70
400	50
500	30
800	20

Solution

Income	Cumulative Income	Cumulative Income (%)	Number of Persons	Cumulative frequency	Cumulative Frequency (%)
0	0	0	0	0	0
100	100	$(100/2000)=5$	80	80	32
200	300	$(300/2000)=15$	70	150	60
400	700	35	50	200	80
500	1200	60	30	230	92
800	2000	100	20	250	100



Practical Problem_3: Lorenz Curve_2

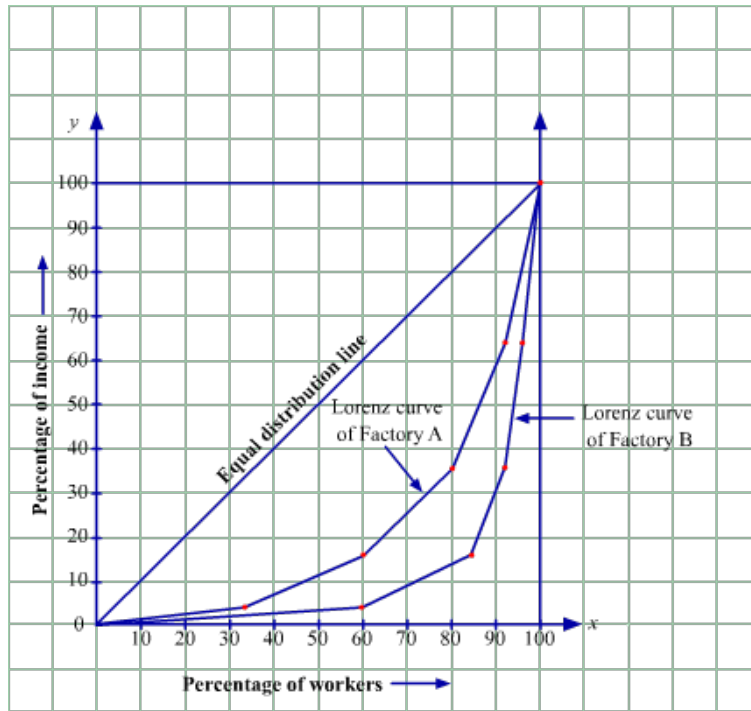
The given table shows the daily income of workers of two factories. Draw the Lorenz Curves for both the factories and comment on your findings.

Daily Income (\$)	0–100	100–200	200–300	300–400	400–500
Factory A	8	7	5	3	2
Factory B	15	6	2	1	1

Solution:

For Factory A						
Daily Income	Mid Value	Cumulative Mid Value	Cumulative Mid Value %	No. of Worker (f)	Cumulative Frequency	Cumulative Frequency %
0–100	50	50	4	8	8	32
100–200	150	200	16	7	15	60
200–300	250	450	36	5	20	80
300–400	350	800	64	3	23	92
400–500	450	1250	100	2	25	100

For Factory B						
Daily Income	Mid Value	Cumulative Mid Value	Cumulative Mid Value of Income (%)	No. of Worker (f)	Cumulative Frequency	Cumulative Frequency of worker (%)
0–100	50	50	4	15	15	60
100–200	150	200	16	6	21	84
200–300	250	450	36	2	23	92
300–400	350	800	64	1	24	96
400–500	450	1250	100	1	25	100

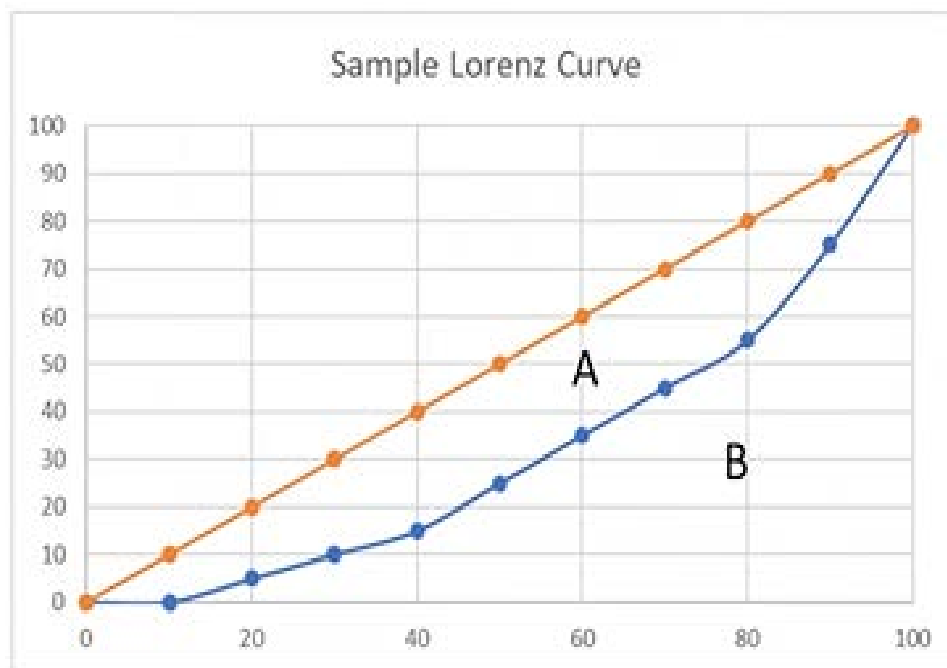


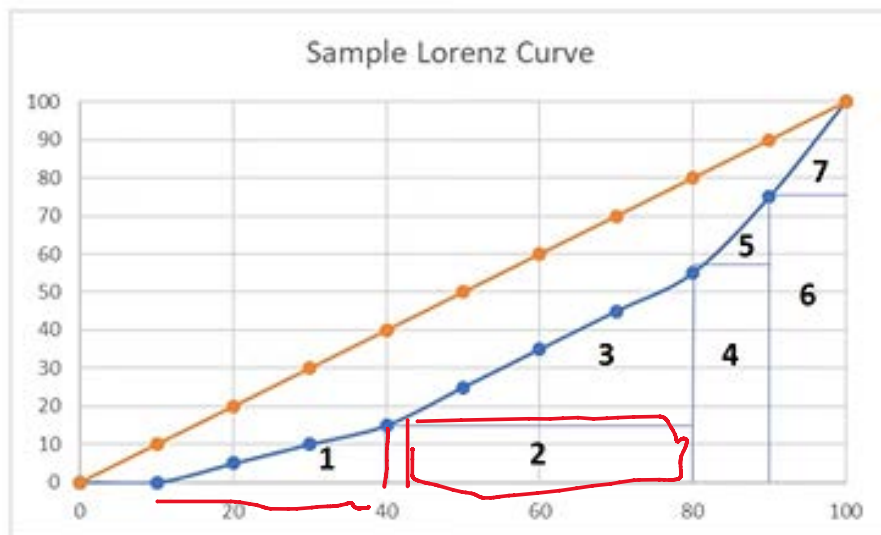
Practical Problem_3: Gini coefficient from Lorenz curve_3

Calculate Gini coefficient from Lorenz curve from the following Table.

Cumulative Population (%)	Cumulative Income (%)
0	0
10	0
20	5
30	10
40	15
50	25
60	35
70	45
80	55
90	75
100	100

Solution:





1:

- Base = $(40 - 10) = 30$
- Height = 15
- Area = $\frac{1}{2} * 30 * 15 = 225$

2:

- Base = $(80 - 40) = 40$
- Height = 15
- Area = $40 * 15 = 600$

3:

- Base = $(80 - 40) = 40$
- Height = $(55 - 15) = 40$
- Area = $\frac{1}{2} * 40 * 40 = 800$

4:

- Base = $(90 - 80) = 10$
- Height = 55
- Area = $55 * 10 = 550$

5:

- Base = $(90 - 80) = 10$
- Height = $(75 - 55) = 20$
- Area = $\frac{1}{2} * 10 * 20 = 100$

6:

- Base = $(100 - 90) = 10$
- Height = 75
- Area = $10 * 75 = 750$

7:

- Base = $(100 - 90) = 10$
- Height = $(100 - 75) = 25$
- Area = $\frac{1}{2} * 10 * 25 = 125$

B:

- Area under the Lorenz Curve
- $1+2+3+4+5+6+7$
- $B = 225+600+800+550+100+750+125$
- $B = 3,150$

A+B:

- Area under the 45-degree line
- $A+B = \frac{1}{2} * 100 * 100$
- $A+B = 5,000$

A:

- Area between the Lorenz Curve and the 45-degree line
- $A = A+B - B$
- $A = 5,000 - 3,150 = 1,850$

Gini coefficient:

- $A/(A+B)$
- Gini = $1,850/5,000$
- **Gini = 0.37**